Linear Regression



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Learning Objectives

- Explain the concept of Linear Regression
- Understand slope and intercept
- Calculate predicted values and residuals
- Explain how a line is fitted to data
- Identify positive, negative, and weak relationships
- Explain Multiple Linear Regression
- Understand the loss/cost function
- Interpret regression coefficients

What is Linear Regression?

A supervised learning algorithm used to predict a continuous numerical value.

Input variable(s) → **X**

Output variable → **Y**

Examples:

- Study hours → Marks;
- House area → House price;
- Experience → Salary;
- Advertising → Sales

The Simple Linear Regression Model

$$y = \beta_0 + \beta_1 x$$

- y: response (dependent variable) x: predictor (independent variable)
- β_0: intercept β_1: slope
- From a sample we estimate the line: $\hat{y} = b_0 + b_1 x$
- $\hat{y}$ ("y-hat") is the predicted value for a given x
- "Linear" means linear in the coefficients: a straight line for one predictor

Fitting a Line

- The data points may not lie exactly on a straight line.
- Linear Regression tries to find a line that:
  - Represents the overall pattern
  - Gives reasonable predictions
  - Minimizes prediction errors

- Goal: Find the line that fits the data as well as possible.

Slope

- Slope tells us how much the predicted Y changes when X increases by 1 unit.
- The slope indicates the steepness of a line

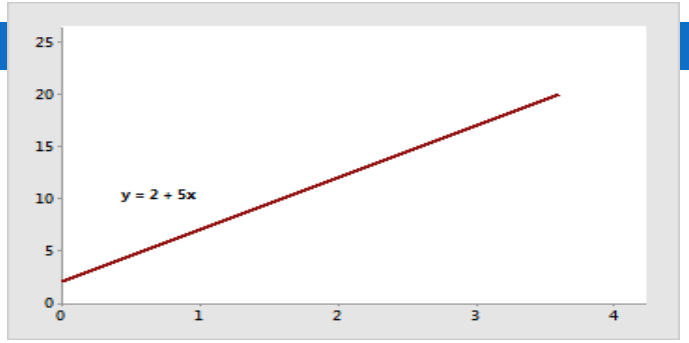
Example: $y = 40 + 5x$

Slope = 5

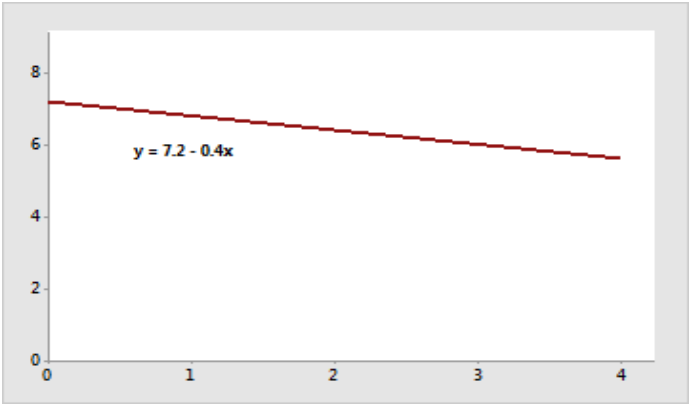
Interpretation: For every 1-unit increase in X, predicted Y increases by 5 units.

Positive and Negative Slope

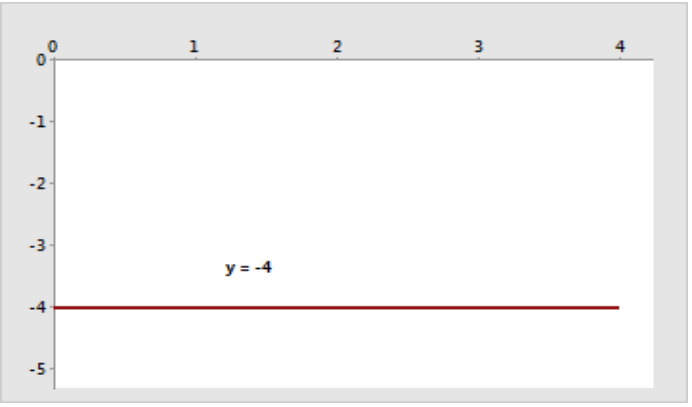
- Positive Linear Relationship: The slope is positive 5. When x increases by 1, y increases by 5. The y-intercept is 2.



- Negative Linear Relationship: The slope is negative 0.4. When x increases by 1, y decreases by 0.4. The y-intercept is 7.2.



- No Relationship: The slope is 0. When x increases by 1, y neither increases or decreases. The y-intercept is -4.



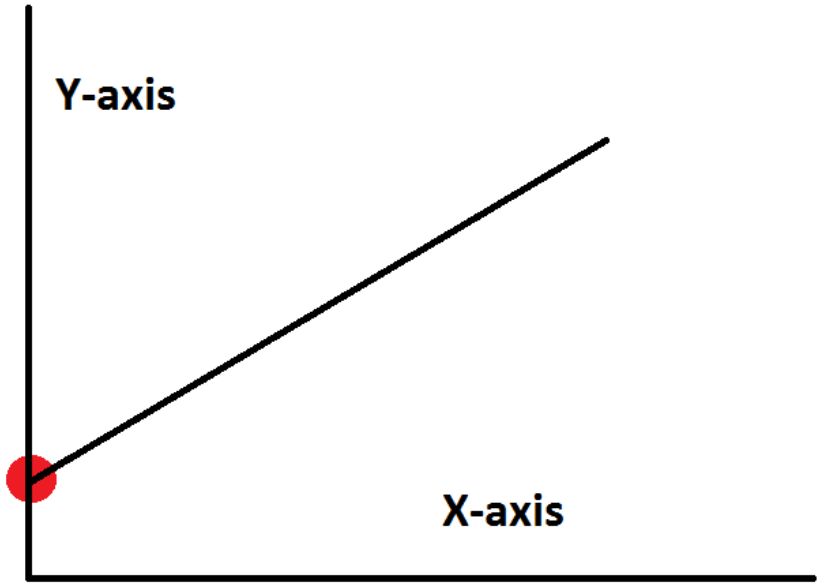
Intercept

- Intercept is the predicted value of Y when X = 0.
- Example: y = 40 + 5x

  When x = 0:

  y = 40

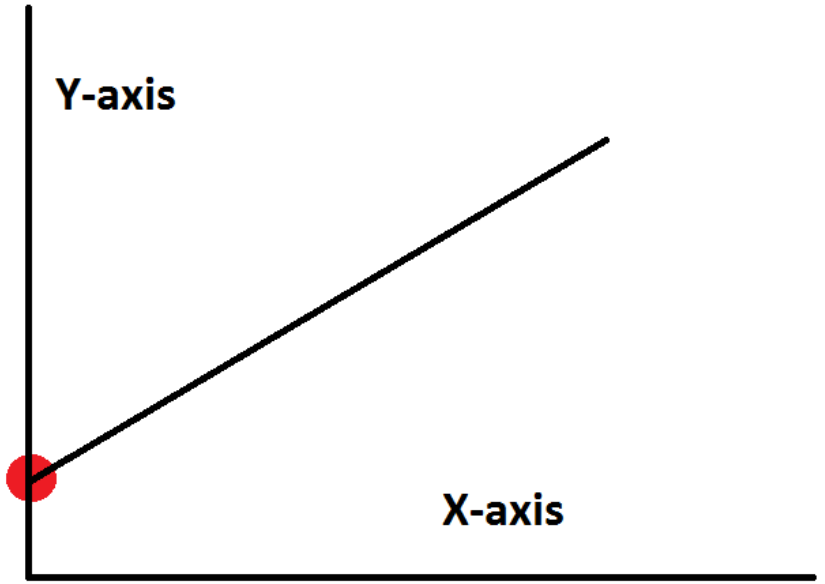
  Intercept = 40



- The slope indicates the steepness of a line and the intercept indicates the location where it intersects an axis. The slope and the intercept define the linear relationship between two variables, and can be used to estimate an average rate of change.

y-intercept

- A positive y-intercept means the line crosses the y-axis above the origin, while a negative y-intercept means that the line crosses below the origin. Simply by changing the values of m and b, we can define any straight line. That's how powerful and versatile the slope intercept formula is.



Making a Prediction

Suppose: $\hat{y} = 40 + 5x$

A student studies for 6 hours.

$\hat{y} = 40 + 5(6)$

$\hat{y} = 70$

Predicted Marks = 70

Actual vs Predicted

Actual Value: The value observed in the real dataset.

Predicted Value: The value produced by the regression model.

Example:

Actual marks = 74

Predicted marks = 70

Difference = 4

This difference is called the **residual**.

Residual

Residual: $e = y - \hat{y}$

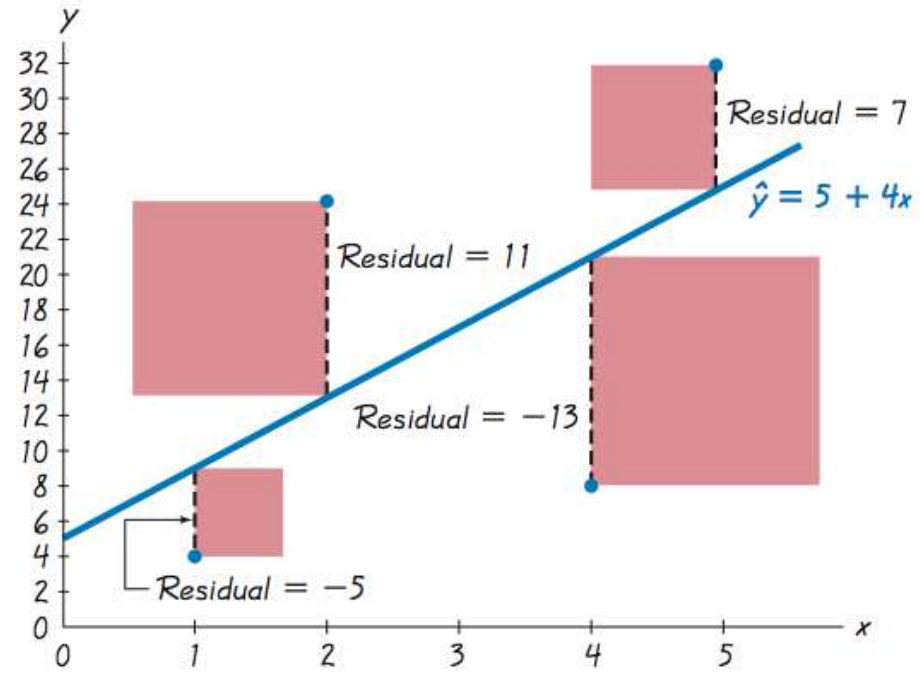
y = actual value

$\hat{y}$ = predicted value

e = residual

Example: $e = 74 - 70 = 4$

Residual = 4



Positive Residual: $y - \hat{y} > 0$

Actual > Predicted → Model underestimated

Negative Residual: $y - \hat{y} < 0$

Actual < Predicted → Model overestimated

Fitting a Line: Least Squares

- Infinitely many lines pass near the data; which is best?
- Least squares: choose b_0, b_1 that minimise SSE = $\Sigma(y_i - \hat{y}_i)^2$
- Squaring stops positive and negative errors cancelling and punishes big misses

$$\hat{y} = b_o + b_1 x$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Example

Data: hours studied x and exam score y for 5 students

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$	$(x-\bar{x})^2$
1	50	-2	-14	28	4
2	55	-1	-9	9	1
3	65	0	1	0	0
4	70	1	6	6	1
5	80	2	16	32	4
Σ				75	10

$\bar{x} = 15/5 = 3$

$\bar{y} = 320/5 = 64$

$b_1 = 75/10 = 7.5$

$b_0 = 64 - 7.5(3) = 41.5$

$\hat{y} = 41.5 + 7.5x$

Interpretation: each extra study hour adds about 7.5 marks; 0 hours predicts 41.5.

Class Activity: Linear Regression

A shop owner wants to predict **monthly sales based on the amount spent on advertising**. The following data was collected for five months:

Month	Advertising Cost (X)	Sales (Y)
1	2	40
2	3	45
3	5	52
4	6	58
5	8	65

Step 1: Calculate the means

$$\bar{X} = \frac{2+3+5+6+8}{5} = \frac{24}{5} = 4.8$$

$$\bar{Y} = \frac{40+45+52+58+65}{5} = \frac{260}{5} = 52$$

$$\boxed{\bar{X} = 4.8, \quad \bar{Y} = 52}$$

Class Activity: Linear Regression

X	Y	$X - \overline{X}$	$Y - \overline{Y}$	$(X - \overline{X})(Y - \overline{Y})$	$(X - \overline{X})^2$
2	40	-2.8	-12	33.6	7.84
3	45	-1.8	-7	12.6	3.24
5	52	0.2	0	0	0.04
6	58	1.2	6	7.2	1.44
8	65	3.2	13	41.6	10.24
Total				95	22.8

$$b_1 = \frac{95}{22.8} = 4.167$$

$$\boxed{b_1 \approx 4.17}$$

Class Activity: Linear Regression

Step 3: Calculate the intercept

$$b_0 = \bar{Y} - b_1\bar{X}$$

$$b_0 = 52 - (4.167)(4.8)$$

$$b_0 \approx 32$$

Step 4: Regression equation

$$\hat{Y} = b_0 + b_1 X$$

$$\boxed{\hat{Y} = 32 + 4.167X}$$

Prediction:
For $X = 7$:

$$\hat{Y} = 32 + 4.167(7)$$

$$\hat{Y} = 32 + 29.169$$

$$\boxed{\hat{Y} \approx 61.17}$$

Predicted sales ≈ 61.17 thousand units.

Class Activity: Multiple Linear Regression

1. The matrix formula

$$\boxed{\mathbf{b} = (X^T X)^{-1} X^T Y}$$

Where:

- X = input matrix containing independent variables
- X^T = transpose of the input matrix
- $(X^T X)^{-1}$ = inverse of the matrix $X^T X$
- Y = output matrix containing actual values
- $\mathbf{b}$ = matrix of regression coefficients

For our example, the coefficients are b_0, b_1, b_2.

Class Activity: Multiple Linear Regression

2. Construct the matrices

X_1	X_2	Y
1	2	10
2	1	11
3	4	20
4	3	21
5	5	26

Step A: Input matrix X

We add a column of 1s for the intercept b_0.

$$X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 3 & 4 \\ 1 & 4 & 3 \\ 1 & 5 & 5 \end{bmatrix}$$

Class Activity: Multiple Linear Regression

Step B: Output matrix Y

$$Y = \begin{bmatrix} 10 \\ 11 \\ 20 \\ 21 \\ 26 \end{bmatrix}$$

Step C: Transpose X

Rows become columns:

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 4 & 3 & 5 \end{bmatrix}$$

Class Activity: Multiple Linear Regression

3. Calculate $X^T X$

Multiply the 3×5 matrix X^T by the 5×3 matrix X:

$$X^T X = \begin{bmatrix} 5 & 15 & 15 \\ 15 & 55 & 53 \\ 15 & 53 & 55 \end{bmatrix}$$

4. Calculate $X^T Y$

Multiply the 3×5 matrix X^T by the 5×1 matrix Y:

$$X^T Y = \begin{bmatrix} 88 \\ 306 \\ 304 \end{bmatrix}$$

Class Activity: Multiple Linear Regression

Therefore, our formula becomes:

$$\mathbf{b} = \begin{bmatrix} 5 & 15 & 15 \\ 15 & 55 & 53 \\ 15 & 53 & 55 \end{bmatrix}^{-1} \begin{bmatrix} 88 \\ 306 \\ 304 \end{bmatrix}$$

5. Calculate the coefficients

After calculating the inverse and performing the matrix multiplication, we obtain:

$$\mathbf{b} = \begin{bmatrix} 3.9333 \\ 2.7778 \\ 1.7778 \end{bmatrix}$$

Class Activity: Multiple Linear Regression

The final regression equation is:

$$\boxed{\hat{Y} = 3.9333 + 2.7778X_1 + 1.7778X_2}$$

This gives the same result as solving the normal equations manually.

Class Activity: Multiple Linear Regression

Study hours (X_1)	Attendance (X_2)	Exam marks (Y)
1	50	45
2	60	50
3	50	55
4	70	65
5	60	70
6	80	80

THANK YOU